Selective Attention Study Report 1

Team DSAC

Abstract

Selective attention governs how individuals filter relevant stimuli from irrelevant ones, and its efficient assessment is critical for both cognitive science and applied settings. This report presents a descriptive analysis of a 2 × 2 mixed factorial study comparing a newly developed mobile "Selective Attention Game" against a standard laboratory visual search task across two target-load conditions (Single vs. Multiple). Using reaction time (RT) and accuracy distributions from 37 participants across 555 lab trials and 546 game-level records, we characterise performance across the four conditions and evaluate four primary hypotheses derived from the study's research questions. Distributions are universally non-normal and right-skewed, consistent with the RT literature. Key findings include: near-identical lab RT across groups despite dramatically divergent game RT, a within-trial slowing effect in the multiple-target lab task consistent with Feature Integration Theory, strong accuracy ceiling effects in both game conditions, and a counterintuitive pattern in which Single Target game RT substantially exceeds Multiple Target game RT. Four primary hypotheses are evaluated descriptively and several additional exploratory hypotheses arising from the data are proposed for Report 2.

Keywords: selective attention, visual search, reaction time, gamified assessment, Feature Integration Theory, descriptive statistics

1. Introduction & Background

Human attention is selective by necessity: the perceptual environment contains far more information than can be processed simultaneously, and the cognitive system must prioritise relevant inputs while suppressing irrelevant ones. This selective filtering process underpins perception, memory encoding, and goal-directed behaviour, and its dysfunction underlies a range of clinical conditions including ADHD and acquired attentional deficits.

The dominant theoretical account of selective attention in visual search is Treisman and Gelade's (1980) Feature Integration Theory (FIT). FIT proposes a two-stage architecture. In the first, pre-attentive stage, basic visual features are processed in parallel across the visual field, a red square "pops out" from white squares instantly. In the second, focused attention stage, features are serially bound into whole objects, requiring attention to be directed to each location in sequence. When multiple targets must be found this serial stage dominates and RT increases with attentional load. This distinction maps directly onto the present study: Single Target (efficient pre-attentive detection) versus Multiple Target (demanding serial localisation of five targets).

Traditional laboratory visual search paradigms are resource-intensive, confined to controlled settings, and can feel monotonous. Gamified cognitive assessment addresses these limitations by embedding tasks within motivating game-like environments. Lumsden et al. (2016) found in a systematic review that gamified formats can maintain data quality comparable to lab tasks while improving engagement but emphasise that each new tool must demonstrate concurrent validity against established measures before applied use. The present study directly answers this call, validating a mobile "Selective Attention Game" against a laboratory visual search task across 37 participants in a 2 × 2 mixed factorial design (Target Load: Single vs. Multiple, between-subjects; Modality: Game vs. Lab, within-subjects). This report is descriptive, characterising the distributions, evaluating the four primary hypotheses, and generating exploratory hypotheses for Report 2.

2. Methods

2.1 Participants

Thirty-seven participants were randomly assigned to Single Target (Group A, n = 21) or Multiple Target (Group

B, n = 16). All completed both the Lab and Game tasks. Approximately half of each group completed the Game first to control for order effects.

2.2 Tasks and Measures

Lab Task (Visual Search). A computer-based task presenting a target (red square) among distractors (white squares). Single Target: one target per trial, one mouse click (RT = time to click). Multiple Target: five targets per trial, participants clicked all five (Primary: Initial RT to first click; Secondary: Inter-Click Interval, ICI, between successive clicks). Fifteen trials per participant, 1:2 target-to-distractor ratio.

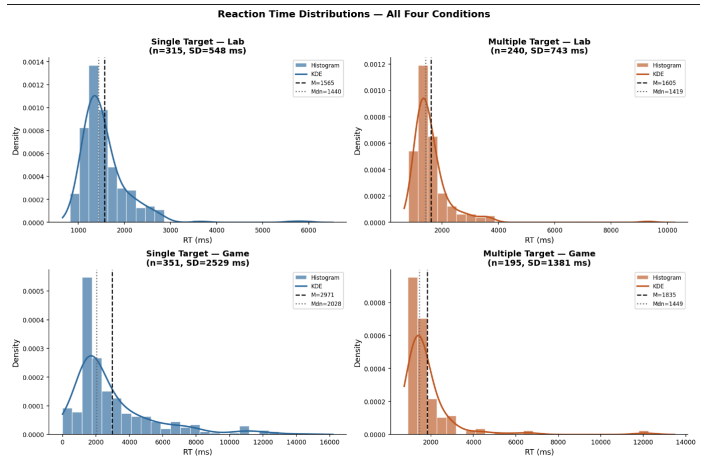
Game Task (Attentional Spotter). A mobile game requiring tapping targets across difficulty levels (1-15 Single; 1-14 Multiple). Per-level measures: Initial RT (ms), Success Rate (%), Hit Rate (%), False Alarms (count), and Multiple group only Average Inter-Target Time (ms). Level completion (pass/fail) recorded.

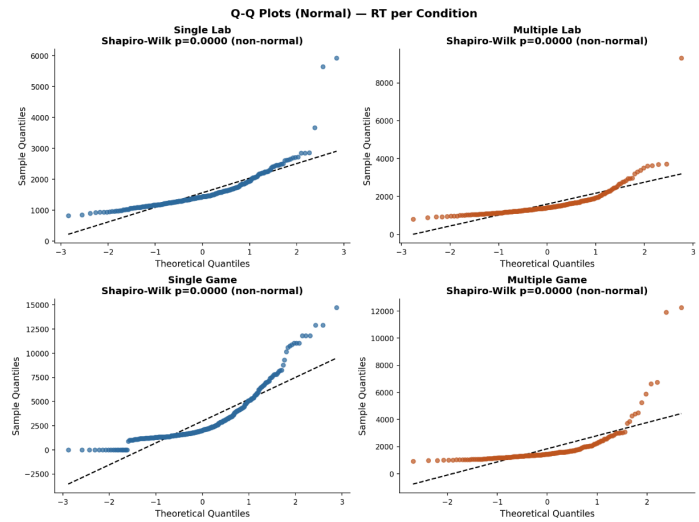
2.3 Analysis Approach

All analyses are descriptive. RT and accuracy are summarised with mean (M), standard deviation (SD), and median (Mdn). Shapiro-Wilk tests confirmed significant non-normality in all four conditions (all $p < .001$), consistent with the right-skewed ex-Gaussian shape typical of RT data (Ratcliff, 1978). Distributions are examined through histograms with KDE, Q-Q plots, violin plots, and scatter plots. Both trial-level and participant-level summaries are provided, with the participant level emphasised as the unit of analysis for Report 2.

3. Dataset Description

The final dataset comprised 555 lab trials and 546 game-level records across all 37 participants, with no missing data. Each participant contributed exactly 15 lab trials (315 Single-group, 240 Multiple-group). In the Game, the Single group contributed 351 level records and the Multiple group 195, the excess reflects level retries (e.g. PID 6: 39 records, PID 5: 19) and the Multiple group reaching lower maximum levels. The figures below provide a first overview of the RT distributions and their departure from normality before the condition-by-condition analysis.





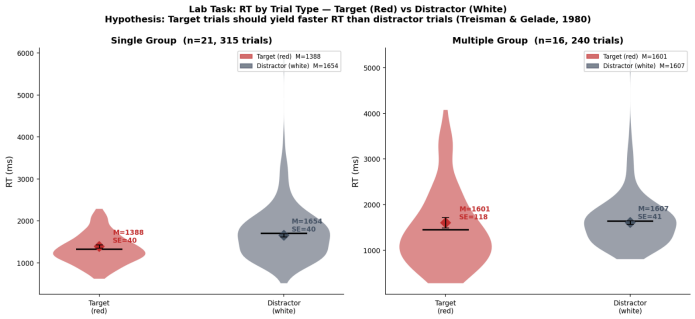
The RT distribution figure immediately shows two features shared across all conditions: strong positive skew and unimodal clustering near the lower end of the RT range. The Single Lab (upper-left) is the most compact, clustering tightly between 1200-1800 ms. The Multiple Lab (upper-right) has a similar mode but a heavier tail (max = 9316 ms). The Single Game (lower-left) is the most extreme, broad and flat, with a tail reaching 14711 ms and an SD nearly as large as the mean. The Multiple Game (lower-right) sits between the two Lab conditions in spread. The Q-Q plots confirm significant non-normality in all four cases: the empirical quantiles track the diagonal in the middle range but curve sharply upward at the upper tail, with the two Game conditions showing the most extreme departure, consistent with the ex-Gaussian shape expected for RT data.

4. Experiments and Observations

4.1 Single Target - Lab Task

315 trials, 21 participants. Overall RT: M = 1565 ms, SD = 548, Mdn = 1440 ms. Distribution positively skewed (skew = 3.49), bulk between 1200-1800 ms, tail extending to 5929 ms (Figure 1, upper-left). At participant level, mean RTs ranged from 1160 ms (PID 17) to 2331 ms (PID 1), a near two-fold spread reflecting genuine individual differences. Trial duration closely mirrored RT (M = 1.61 s, SD = 0.55).

Breaking down by trial type: target-present (red) trials M = 1388 ms (SD = 414, Mdn = 1291 ms) versus distractor-only (white) trials M = 1654 ms (SD = 585, Mdn = 1487 ms), a 266 ms difference. FIT predicts that on distractor-only trials, participants must exhaustively confirm the absence of a target before responding, producing slower and more variable RTs.

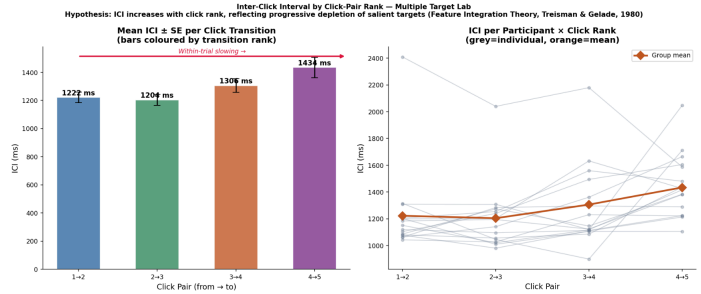


The violin confirms the pattern for the Single group: the red distribution is compact and low, the white distribution has a pronounced upper tail. The same figure shows the Multiple group comparison for completeness discussed further in Section 4.2.

4.2 Multiple Target - Lab Task

240 trials, 16 participants. Initial RT: M = 1605 ms, SD = 743, Mdn = 1419 ms, comparable in mean to Single Lab but more variable (SD = 743 vs. 548) with a heavier tail (max = 9316 ms, skew = 5.38). See Figure 1, upper-right. Total trial duration far longer (M = 6.66 s vs. 1.61 s), expected for five-target trials.

ICI across 904 observations: M = 1290 ms, SD = 756, Mdn = 1100 ms. By click-pair rank: 1→2: 1222 ms, 2→3: 1204 ms, 3→4: 1306 ms, 4→5: 1434 ms, a progressive increase of 212 ms across the sequence.



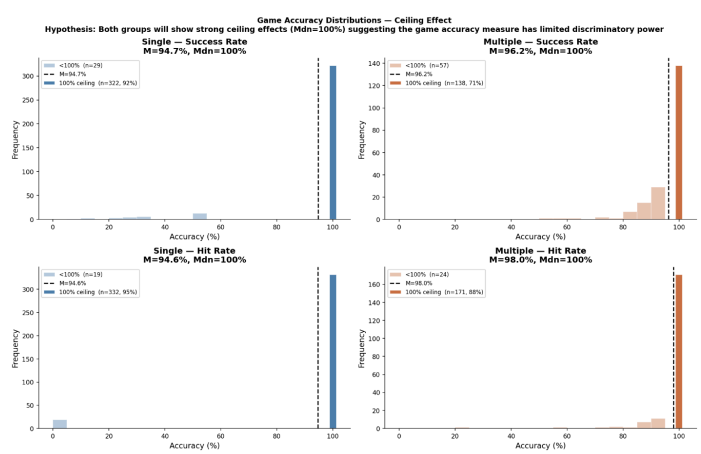
The bar chart and spaghetti plot (Figure 4) confirm the monotonic rank increase, consistent across individual participants. ICI on red trials (M = 1137 ms) was 230 ms shorter than on white distractor trials (M = 1367 ms), consistent with colour-guided localisation being faster when all targets share a salient colour. The Multiple group red vs white violin (Figure 3, right panel) shows near-identical means for initial RT (1601 vs. 1607 ms) but dramatically different variance (SD = 1059 red vs. 523 white) a qualitatively different pattern from the Single group, suggesting colour salience operates differently under multiple-target search demands.

At participant level, initial RTs ranged from 1269 ms (PID 29) to 2129 ms (PID 37).

4.3 Single Target - Game Task

351 level records, 21 participants; all reached Level 15 (completion rate: 94.6%, 332/351). RT: M = 2971 ms, SD = 2529, Mdn = 2028 ms. SD nearly as large as the mean; max = 14711 ms. See Figure 1, lower-left, and Q-Q lower-left for the extreme heavy-tail departure.

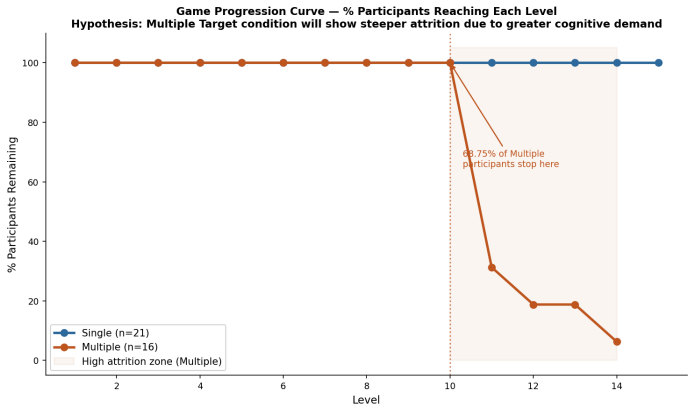
Accuracy showed strong ceiling effects: Success Rate M = 94.7% (SD = 18.1, Mdn = 100%), Hit Rate M = 94.6% (SD = 22.7, Mdn = 100%), False Alarms M = 0.21 (Mdn = 0).

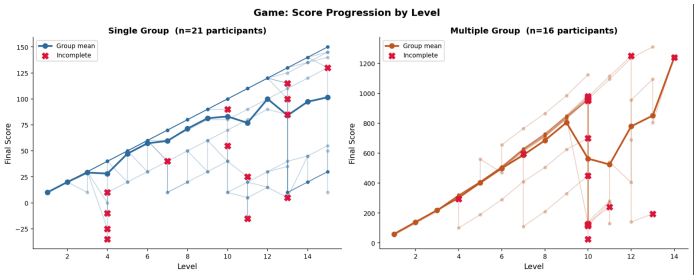


The ceiling plot (Figure 5) shows the vast majority of level records at exactly 100%; the non-zero means are driven by a small number of outlier levels. Negative skew (SR skew = −3.31) confirms this ceiling-dominated structure.

4.4 Multiple Target - Game Task

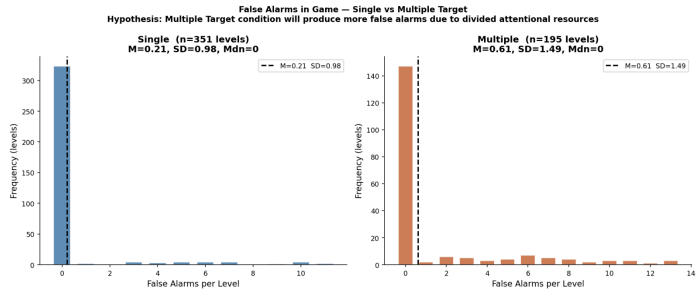
195 level records, 16 participants. Critical observation: 11 of 16 participants (68.75%) reached a maximum of Level 10 only; 5 progressed to Levels 11-14. Mean max level = 10.75 (SD = 1.34) vs. 15.00 for Single. Completion rate: 87.7% (171/195), 24 failed attempts.





The progression curve (Figure 7) shows both groups at 100% through Level 10, after which the Multiple group drops sharply to 31.25% at Level 11 and 6.25% at Level 14. Score progression shows the Multiple group's scores climbing steeply but with increasing variance and failure markers multiplying from Level 10 onward. Data above Level 10 for the Multiple group come from at most 5 participants and are not representative. Because of the sharp reduction in observations above Level 10, mean (average) values at higher levels are unstable and should be interpreted cautiously. With very few observations, averages become highly sensitive to slow outlier trials and do not reliably represent the underlying performance distribution. For these upper levels, more robust approaches such as resampling or sampling-balanced analyses may be required in future analyses to obtain stable estimates.

RT: M = 1835 ms, SD = 1381, Mdn = 1449 ms. Average Inter-Target Time: M = 1111 ms (SD = 566), closely paralleling Lab ICI (M = 1290 ms). Accuracy ceiling maintained: SR M = 96.2% (Mdn = 100%), HR M = 98.0% (Mdn = 100%) see Figure 6, Multiple panels.



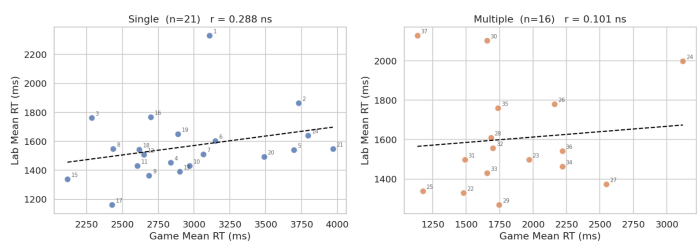
False Alarms: Multiple M = 0.61 (SD = 1.49, max = 13) vs. Single M = 0.21 (SD = 0.98). Both zero-inflated (Mdn = 0) but the Multiple tail is substantially heavier, extending to 13 false alarms on a single level.

5. Primary Hypotheses

The four primary hypotheses correspond directly to the study's four research questions.

H1 - Concurrent Validity: Game and Lab RT Will Correlate Positively Within Each Group (RQ1)

Hypothesis. If the Game validly measures the same attentional construct as the Lab, participants who respond quickly in the Lab should also respond quickly in the Game. We hypothesise a significant positive correlation between participant-mean Lab RT and participant-mean Game RT within each group.



Results. Single group (n = 21): r = 0.144, p = .534. Multiple group (n = 15): r = 0.066, p = .815. Both scatter plots show diffuse, near-horizontal clouds with no discernible upward trend.

Interpretation. H1 is not supported descriptively. The null hypothesis of no correlation is retained in both groups. However, small sample sizes give low statistical

power (with n = 21, a correlation of r = .43 is needed to reach p = .05), and the moderate Lab reliability (α = .744 Single, α = .621 Multiple) places a theoretical ceiling on the maximum observable correlation and even a perfectly reliable Game measure could not correlate with the Single Lab above √.744 ≈ .86. Platform differences (mouse vs. touchscreen) and structural differences in how RT is generated across modalities further constrain validity correlations. Report 2 will evaluate this formally with confidence intervals.

H2 - Target Load Effect: Multiple Target Performance Will Be Worse Than Single Target (RQ2)

Hypothesis. Based on FIT, Multiple Target imposes greater attentional demand. We hypothesise Multiple Target performance will be significantly worse, i.e. higher RT, lower accuracy, and fewer levels reached, than Single Target.

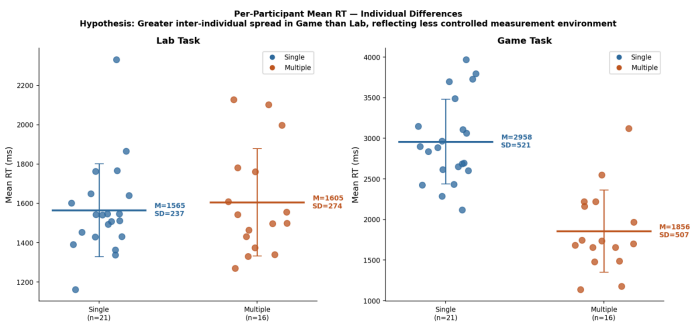


Table 1. Participant-Level Mean RT (ms) - Summary by Condition

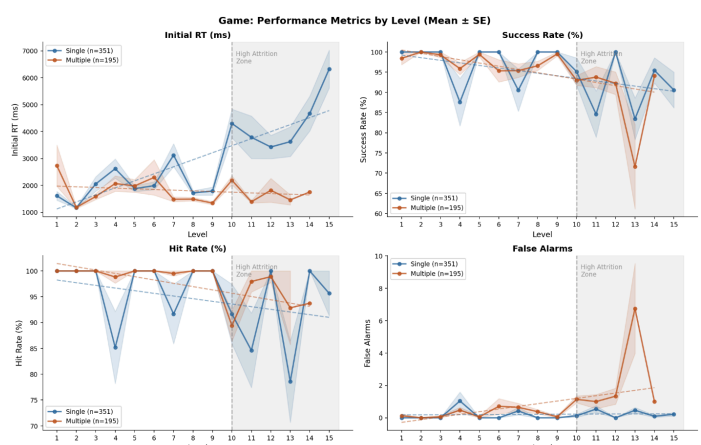
Group	Modality	n	M (ms)	SD	Mdn (ms)
Single	Lab	21	1565	237	1538
Single	Game	21	2958	521	2888
Multiple	Lab	16	1605	274	1542
Multiple	Game	16	1856	507	1702

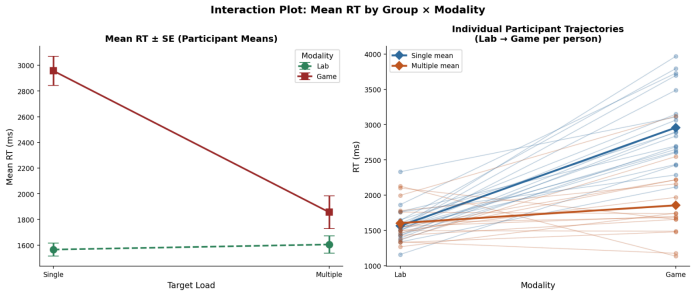
Results. In the Lab, both groups are nearly indistinguishable (1565 vs. 1605 ms, overlapping clusters in Figure 11). In the Game, Single RT (2958 ms) is paradoxically *higher* than Multiple RT (1856 ms). However, Multiple participants reached only 10.75 average max level vs. 15.00 for Single, produced nearly 3× more False Alarms (0.61 vs. 0.21), and showed steeper accuracy decline with level.

Interpretation. H2 is partially supported. Multiple Target performance is demonstrably worse on accuracy, level progression, and error rate which are all consistent with greater attentional demand. The counterintuitive RT reversal in the Game is explained by task structure: in the Multiple game, several targets are simultaneously visible and spatially distributed, making each tap a direct response to a nearby visible target; in the Single game, participants must exhaustively search the entire screen for a single hidden target, inflating RT far beyond what cognitive difficulty alone predicts. This structural confound will be explicitly addressed in Report 2.

H3 - Modality Effect: Game RT Will Differ From Lab RT, With an Interaction by Target Load (RQ3)

Hypothesis. The gamified interface introduces uncontrolled elements absent in the lab (random target placement, touchscreen demands, mobile environment variability). We hypothesise Game RT will be significantly higher than Lab RT, and that this inflation will be larger for the Single group than the Multiple group producing a significant Modality × Target Load interaction.





Results. Single Game-Lab difference: +1393 ms (+89%). Multiple Game-Lab difference: +251 ms (+16%). The interaction plot (Figure 13, left) shows a steep positive slope for Single and a near-flat slope for Multiple, a ratio of approximately 5.5:1. The right panel confirms this at the individual level: virtually every Single participant's line rises steeply from Lab to Game, while Multiple participants' lines are mostly shallow. Game conditions also show substantially larger SD than Lab (Single Game SD = 521 vs. Lab SD = 237; Multiple Game SD = 507 vs. Lab SD = 274).

Interpretation. H3 is strongly supported descriptively. The gamified interface inflates RT substantially, and critically this inflation is not uniform (approximately 5.5 times larger for Single than Multiple). This asymmetry, combined with the near-identical Lab RTs, is the central descriptive finding of the dataset and will form the 2 × 2 Mixed ANOVA interaction test in Report 2. Game conditions also produce more variable performance across participants, consistent with the less controlled environment introducing target-location randomness and device variability as additional noise sources.

H4 - Level Effect: RT Will Increase and Accuracy Will Decrease With Game Level (RQ4)

Hypothesis. Game levels were designed to increase in difficulty. We hypothesise that RT will increase and accuracy measures (SR, HR) will decrease significantly with level in both game conditions, and False Alarms will increase with level.

Results. From Figure 12: Single group shows a clear upward RT trend, particularly steep above Level 7, while SR and HR remain near ceiling at early levels but decline at higher ones. Score progression (Figure 8) increases roughly linearly. Multiple groups show SR and HR declining from Level 5 and False Alarms spiking sharply above Level 10, though data there come from only 5 participants. RT shows no clear trend in the Multiple group before Level 10, possibly because Initial RT in the Multiple game is structurally lower and less sensitive to level scaling.

Interpretation. H4 is descriptively supported for the Single group across all metrics. For the Multiple group, accuracy decline with level is visible but the 68.75% attrition at Level 10 prevents reliable interpretation of upper-level trends. Spearman rank correlations in Report 2 will formally test the monotonic level effects for both groups.

6. Additional Exploratory Hypotheses

Beyond the four primary hypotheses, the descriptive data reveal several additional patterns that motivate exploratory hypotheses for Report 2.

6.1 Search Termination Effect (Red vs White RT)

In the Single Lab, red trials (M = 1388 ms) were 266 ms faster than white trials (M = 1654 ms), consistent with FIT's search termination account; distractor-only trials require exhaustive confirmation of target absence before response. In the Multiple Lab, initial RT was near-identical for red and white trials in mean (1601 vs. 1607 ms) but dramatically different in variance (SD = 1059 vs. 523), suggesting colour salience affects the variability rather than the speed of first-target detection when five targets must be found.

H_E1 - Search Termination: *Distractor-only (white) trials will produce significantly longer RT than target-present (red) trials in the Single condition; this difference will be absent or qualitatively different in the Multiple condition.* Supported descriptively for Single (Δ = 266 ms). The Multiple group pattern i.e. same mean, very different SD, is itself a hypothesis: red trials in the Multiple condition may produce a bimodal RT distribution (fast when the first target is in a salient position, very slow when it is not), to be examined in Report 2.

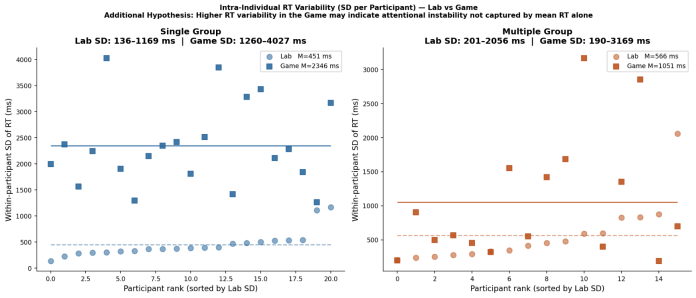
6.2 Within-Trial ICI Slowing and Red vs White ICI

ICI increased progressively from 1222 ms (1→2) to 1434 ms (4→5), consistent with FIT's prediction that the most salient targets are found first and remaining targets require increasingly effortful serial search. ICI was additionally shorter on red trials (M = 1137 ms) than white trials (M = 1367 ms, Δ = 230 ms), consistent with colour-guided

within-trial localisation being faster when all targets share a salient colour.

H_E2 - ICI Slowing and Colour Guidance: *ICI will increase significantly across click rank (formally testable via Kruskal-Wallis in Report 2), and ICI will be significantly shorter on red than white trials (Mann-Whitney U test). Both effects will be consistent with FIT's pre-attentive colour processing mechanism.*

6.3 RT Variability as an Attentional Stability Index



Every participant in both groups shows substantially higher RT variability in the Game than the Lab. Single group: Lab SD M = 451 ms (range 136-1169) vs. Game SD M = 2346 ms (range 1260-4027), more than five times larger. Multiple group: Lab SD M = 566 ms (range 201-2056) vs. Game SD M = 1051 ms (range 190-3169), approximately double. This gap is consistent across every participant rank, not driven by outliers.

Reaction time variability (RTV) is an established index of attentional stability and cognitive control. Stuss et al. (2003) demonstrated that frontal lobe patients show disproportionately elevated RTV even when their mean RT is normal. West et al. (2002) showed RTV correlates with attentional lapses independently of mean speed. The universal Game > Lab RTV suggests the game imposes greater sustained attentional demands, consistent with the unpredictable target timing and spatial randomness of the mobile format.

H_E3 - RTV Predicts Game Accuracy: *Participants with higher within-session Game RTV will show lower Success Rates and more False Alarms, reflecting attentional instability that manifests as both occasional very slow responses and occasional errors. This relationship is expected to be weaker in the Lab where the controlled format provides external attentional scaffolding. If confirmed, RTV would constitute a more sensitive validity indicator than mean RT alone.*

6.4 Accuracy Ceiling as a Validity Constraint

Both game conditions show Mdn = 100% for SR and HR (Figure 6), leaving near-zero variance in accuracy scores.

H_E4 - Accuracy Ceiling Limits Validity: *Because accuracy is ceiling-constrained, RT will show stronger concurrent validity with the Lab than SR or HR in Report 2. False Alarms, which are not ceiling-constrained and differ meaningfully across groups (0.21 vs. 0.61), may be a more discriminating accuracy-adjacent measure and should be tested as an alternative validity criterion.*

7. Key Questions Answered

Q1: Do Lab and Game RTs correlate within each group (concurrent validity)? A: Not supported descriptively. Single: r = .144, p = .534. Multiple: r = .066, p = .815. Both scatters show diffuse, near-horizontal clouds. However, low power (n = 21 and n = 15) and moderate Lab reliability (α = .744 and .621) constrain what correlations are realistically detectable. Report 2 will provide formal tests with confidence intervals.

Q2: Is performance worse in the Multiple Target condition? A: Partially. Multiple participants reached lower max game levels (10.75 vs. 15.00), produced nearly 3× more False Alarms (0.61 vs. 0.21), and showed steeper accuracy decline with levels which are all consistent with greater attentional demand. However, Multiple Game Initial RT was paradoxically lower than Single Game RT, explained by structural task differences rather than a performance advantage.

Q3: Does the gamified interface alter performance vs. the Lab? A: Yes, and differentially by group. Single Game RT nearly doubled Lab RT (+1393 ms, +89%). Multiple Game RT showed only a modest increase (+251 ms, +16%). The resulting 5.5:1 asymmetry (despite near-identical Lab RTs) is the central descriptive finding and previews a significant Modality × Load interaction in Report 2's mixed ANOVA.

Q4: Do game levels scale in difficulty? A: Yes, most clearly for the Single group where all 21 participants provide data at all 15 levels: RT rises and accuracy declines with level. Multiple groups show the same accuracy trends but data above Level 10 come from only 5 of 16 participants, making those trends unreliable.

Q5: Does target colour (red vs white) affect RT? A: Yes, in the Single Lab red trials were 266 ms faster than white (1388 vs. 1654 ms), consistent with search termination. In the Multiple Lab, ICI on red trials was 230 ms shorter than white (1137 vs. 1367 ms), consistent with colour-guided within-trial search. The Multiple Lab initial RT red/white difference is near-zero in mean but large in variance asymmetry which is a distinct pattern worth further investigation.

Q6: Is there a within-trial slowing effect in the Multiple Lab ICI? A: Yes. Mean ICI increased from 1222 ms (1→2) to 1434 ms (4→5), consistent with FIT's prediction that the most salient targets are detected first, leaving progressively harder targets for later clicks.

Q7: Are the observed effects practically meaningful? A: The Single Game-Lab RT difference (+1393 ms, +89%) is large and practically significant. The Target Load difference in False Alarms (0.61 vs. 0.21) and Max Level (10.75 vs. 15.00) are also meaningful. The Lab RT group difference (40 ms, 2.6%) is trivially small. Most accuracy measures are ceiling-constrained and therefore practically uninformative as individual difference measures.

8. Reliability of the Lab Measure

Split-half reliability (odd vs. even trials, Spearman-Brown corrected): Single $r = .503$, Cronbach's $\alpha = .744$ (good). Multiple $r = .386$, $\alpha = .621$ (acceptable). These values place a theoretical ceiling on the maximum observable concurrent validity correlation in Report 2 and even a perfectly reliable game measure could not correlate with the Single Lab above $\sqrt{.744} \approx .86$, and in practice much lower once game reliability is factored in. Any near-zero validity correlation in Report 2 must therefore be evaluated against this ceiling rather than treated as definitive evidence of invalidity.

9. Conclusion and Future Directions

This report characterised the RT and accuracy distributions of 37 participants across four conditions of a 2 × 2 mixed factorial selective attention study and evaluated four primary hypotheses descriptively. H1 (concurrent validity) was not supported, though low power and moderate reliability constrain interpretation. H2 (target load effect) was partially supported where Multiple Target performance was worse on accuracy, level progression, and false alarms, but not on Initial RT due to a task-structural confound. H3 (modality effect and interaction) was strongly supported where the gamified interface inflated RT by +1393 ms for Single but only +251 ms for Multiple, a 5.5:1 asymmetry that will be the central formal test in Report 2. H4 (level scaling) was descriptively supported for the Single group and partially for Multiple, with the 68.75% attrition at Level 10 limiting upper-level interpretation.

Four exploratory hypotheses were additionally proposed: H_E1 (search termination and red/white RT pattern), H_E2 (ICI slowing and colour-guided localisation), H_E3 (RTV predicts game accuracy), and H_E4 (accuracy ceiling constrains validity).

Future directions: (1) formally test H3's interaction using a 2 × 2 Mixed ANOVA in Report 2; (2) test H2's load effect using independent and paired t-tests; (3) formally test H4's level scaling using Spearman correlations; (4) evaluate H_E3 by correlating within-participant Game RTV with Success Rate and False Alarms; (5) test H_E4 by comparing validity correlations for RT versus accuracy measures; (6) recruit a larger Multiple Target sample to enable level analyses above Level 10; (7) establish test-retest reliability of the Game across sessions, which Lumsden et al. (2016) identify as a prerequisite for applied or clinical deployment; (8) when running ANOVA, conduct appropriate diagnostic checks and, given the observed non-normality of RT distributions, consider robust or non-parametric ANOVA alternatives where assumptions are violated; (9) retain and analyse trial-level data, particularly for the Game task; (10) improve the concurrent validity analysis by correlating Lab RT with Level 1 Game RT, which most closely matches the single-trial structure of the lab task; (11) examine progression using mixed-effects modelling with participant-level observations; and (12) model reaction time using log-transformed RT and estimate separate mixed-effects regressions for the Single and Multiple groups, with modality (Lab vs. Phone) as the predictor.

Repository: Github Link

Contributions:
Shravan- Data Preprocessing
Masumi- Visualisations
Tanish- Hypotheses and EDA

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